ST ANDREW'S JUNIOR COLLEGE
JC2 Preliminary Examination (2008)
H1 Biology
Paper 1 & 2: MCQ, STQ & Essay

MARK SCHEME

Disclaimer
This mark scheme should be used purely as a revision tool.
Assumptions about future mark schemes
on the basis of one year's document should be avoided;
whilst the guiding principles of assessment remain constant,
details will change, depending on the content of a particular examination paper.

Key to mark scheme

ref.	with reference to
/	alternative wordings/phrases
(brackets)	recommended wording that is optional
underline	accept this word only
OVP	other valid points

Paper 1: MCQ

Q1	C	**Q2**	B	**Q3**	B	**Q4**	C	**Q5**	A
Q6	B	**Q7**	D	**Q8**	B	**Q9**	D	**Q10**	C
Q11	B	**Q12**	A	**Q13**	C	**Q14**	D	**Q15**	B
Q16	D	**Q17**	D	**Q18**	A	**Q19**	C	**Q20**	C
Q21	D	**Q22**	A	**Q23**	A	**Q24**	C	**Q25**	C
Q26	C	**Q27**	D	**Q28**	C	**Q29**	D	**Q30**	A

Paper 2: STQ & Essay

Section A

QUESTION 1

The Figure 1.1 below shows a model of the enzyme chymotrypsin. Chymotrypsin catalyses the hydrolysis of peptide bonds of proteins. The active site of chymotrypsin is located in a slight depression on one side of the molecule. Three amino acids form the active site. These three amino acids are some distance apart on the polypeptide chain but close together in the active site.

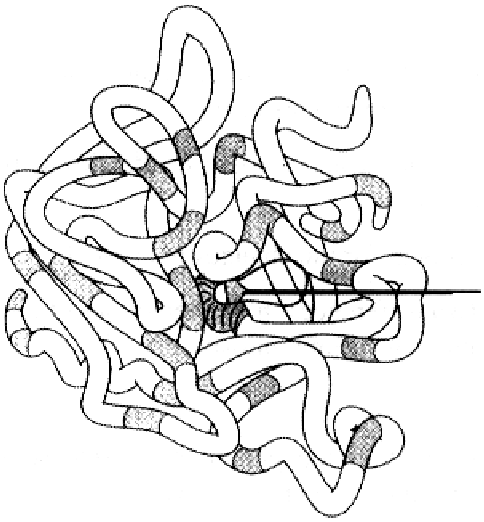


Figure 1.1

(a) Describe how amino acid residues at different positions in the protein may be brought together in the active site when the enzyme is synthesised. [2]

1 brought close together in the active site by the <u>folding</u> of polypeptide chain

2 ref. (different chemical nature of) R groups eg. ionic / polar / non-polar / acidic / basic (at least 2 examples);

3 positions maintained by <u>hydrophobic interactions, hydrogen bonds, ionic bonds</u> and <u>disulphide bonds</u> (at least 2 examples)

(b) Explain how a substrate may be attached to the enzyme. [3]

1 lock-and-key-hypothesis where the substrate referred to as <u>key</u> while enzyme is referred to a <u>lock/padlock</u> as the active sites complementary to substrate

2 induced-fit hypothesis where the initial shape of active site of enzyme <u>might not</u> be complementary to shape of substrate;

3 binding of substrate to enzyme causes an alteration in the conformation of <u>active site</u> to enabling better fit / more effective binding of substrate

4 ref. successful/effective enzyme-substrate collision / ref. correct orientation between active site and substrate

The following figure 1.2 shows the effects of increasing substrate concentration on the rate of an enzyme-catalysed reaction at a temperature of 35°C and a constant pH.

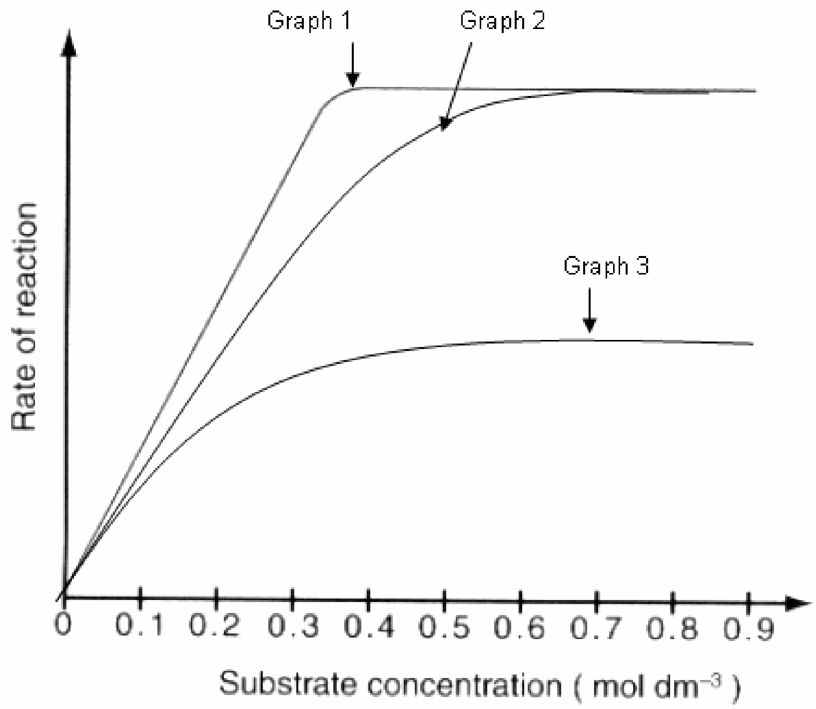


Figure 1.2

(c) With reference to graph 1, explain why an increase in substrate concentration at low substrate concentrations increases the rate of reaction but an increase at high substrate concentrations does not have the same effect. [4]

<u>At low substrate concentration</u>

1 substrate conc. increases, rate of reaction increases, ref. figures 0 to 0.37 mol dm^{-3};

2 increase in frequency of <u>effective/successful</u> collisions between enzyme and substrate molecules, more enzyme-substrate complexes formed;

3 ref. enzyme is in excess / substrate is limiting

<u>At high substrate concentration</u>

4 further increase in substrate conc., no further increase in rate of reaction, ref. figures 0.37 to 0.9 mol dm^{-3} / beyond 0.37 mol dm^{-3}

5 all available active sites (becoming) occupied / enzyme (becoming) saturated with substrate;

6 ref. substrate is no longer limiting / enzyme concentration is limiting;

(d) Complete the table below to show 3 differences between inhibitors in Graph 2 and Graph 3 and their mode of action. [3]

Factor of comparison	Graph 2 : Competitive Inhibition	Graph 3 : Non-competitive Inhibition
Structural resemblance to the substrate	Inhibitor is structurally similar to the substrate	Inhibitor is not structurally similar to the substrate
Mode of action on the enzyme	Competitive inhibitors bind reversibly to the active site of the enzyme Or Competitive inhibitors <u>compete with substrates for active site</u> / inhibitors do not render the enzymes as non-functional and can be re-used	Non-competitive inhibitors bind to a region other than the active site of the enzyme i.e. allosteric site Or Non-competitive inhibitors <u>cause a conformational change of the active site</u> after binding to the allosteric site / renders the enzyme as non-functional
Effect of increasing substrate concentration on action	Inhibition can be overcome Or The rate of reaction reaches the same V_{max} as that catalyzed by uninhibited enzyme at high substrate concentration	Inhibition cannot be overcome. Or The rate of reaction does not reach the same V_{max} as that catalyzed by uninhibited enzyme at high substrate concentration

[Q1 Total: 12]

QUESTION 2

Silena alba (White Champion), there are separate male and female plants. Male plants are XY and female plants are XX.

In 1912, Baur described the first example of sex-linkage in plants. He discovered an unusual form of *Silena alba* with narrow leaves. It was found to be a male plant. Pollen from this plant was used to fertilise a female plant with normal broad leaves, and all progeny, male and female, had normal broad leaves. When these normal broad-leaved plants were cross fertilised, they produces 167 plants with broad leaves and 60 with narrow leaves.

When the narrow-leaved plants were grown to maturity all were male, while of the broad-leaved plants, 56 were male and 111 were female.

(a) Draw a genetic diagram to explain both crosses. [5]

	Female	Male
Parental phenotypes:	Broad leaves	Narrow leaves
Parental genotypes:	$X^B X^B$	$X^b Y$
Parental gametes:	X^B	X^b Y

F_1 genotypes: $X^B X^b$ $X^B Y$
F_1 phenotypes: broad leaves, female broad leaves, male

Therefore X^B is the dominant allele for broad leaves, X^b and the recessive allele for narrow leaves

F_1 genotypes: $X^B X^b$ $X^B Y$
F_1 phenotypes: broad leaves, female broad leaves, male

F_1 gametes:	X^B X^b	X^B Y

Punnett square showing F_2 genotypes

	X^B	X^b
X^B	X^BX^B	X^BX^b
Y	X^BY	X^bY

F_2 genotypes	X^BX^B , X^BX^b	X^BY	X^bY
F_2 phenotypes	Broad leaves, female	Broad leaves, male	Narrow leaves, male
F_2 phenotypic ratio	2	1	1

F_2 phenotypes	Broad leaves, female Broad leaves, male	Narrow leaves, male
F_2 phenotypic ratio	3	1

1 Parental gametes – X^B , X^b and Y
2 F1 genotypes – X^BX^b and X^BY
3 F1 phenotype –broad leaves, female and broad leaves, male
4 F1 parental gametes – X^B, X^b and Y
5 Punnett square / F2 genotypes corresponds genotypes to phenotypes / legend
6 F2 phenotypic ratio – , 3:1; 2:1:1

(b) Explain why there were no female narrow-leaved plants. [2]

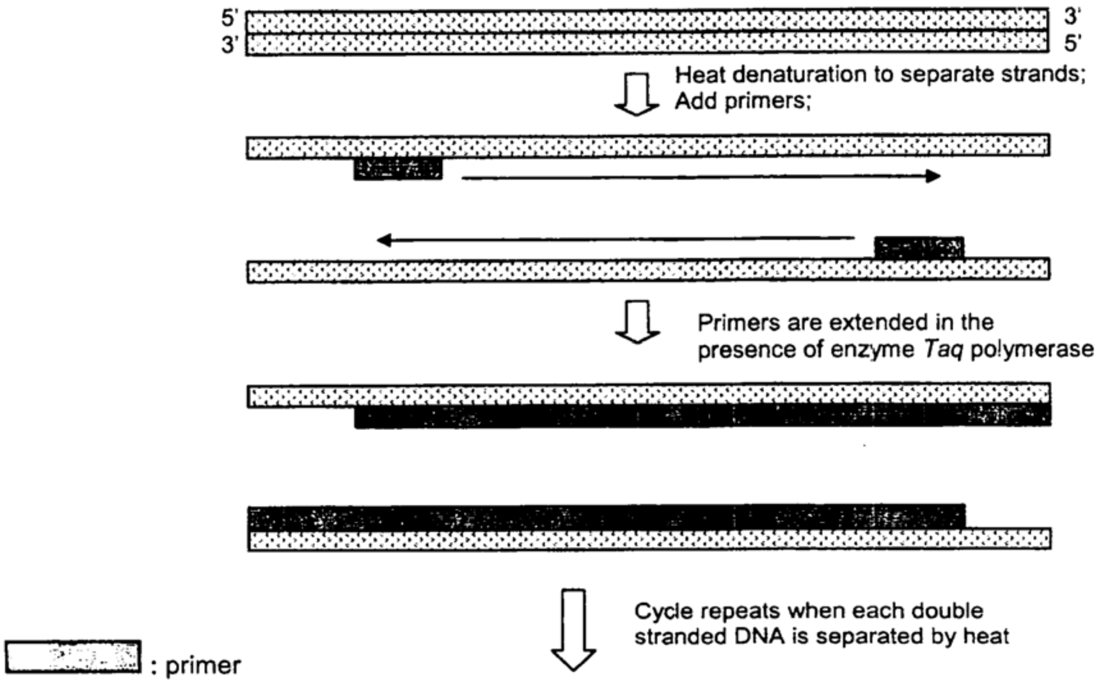
1 requires both recessive allele for narrow leaves for phenotype to be expressed
2 always inherits dominant allele from male parent

(c) Suggest how a female narrow-leaved plant could be produced. [1]

1 Cross heterozygous broad leaved female with narrow leaved male plant

[Q2 Total: 8]

QUESTION 3
The polymerase chain reaction is a process used to make many copies of a piece of DNA. The process is outline in the figure below.



(a) The primer is a short sequence of nucleotides that binds to the DNA. Explain why the primer only binds to the DNA at a particular position.

1 Has complementary base sequence to the DNA at that position;

(b) Describe the role of the primer in the PCR.

1 Allows DNA polymerase to bind and add deoxyribonucleotide to the 3'OH end of the primer;

(c) Both processes, polymerase chain reaction and DNA semi-conservative replication involves elongation of polynucleotide chains. List 2 differences between the polymerase chain reaction and DNA semi-conservative replication.

1 PCR requires only two types of primers;
2 Semi-conservative replication requires the use of many primers for the synthesis of the lagging strand;

3 PCR uses heat to separate the DNA strands
4 The helicase is required to unwind and separate the DNA strands in semi-conservative replication;

5 Replication in PCR is continuous for both strands
6 Whereas there a discontinuous synthesis on the lagging strand while the leading strand is synthesized continuously during semi-conservative replication;

The cloned DNA and plasmids were first cut by the same restriction enzyme and then mixed together with DNA ligase. Several types of plasmids were formed. Some contained the human DNA in the centre of the gene coding for resistance to tetracycline. The different types of plasmids are shown in Figure 3.2.

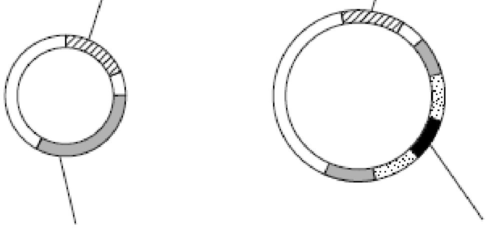


Figure 3.2

(e) What are the properties of restriction enzymes that make them useful in genetic engineering? [2]

1 Recognizes a specific sequence of bases (restriction site) on the DNA molecule and cuts at this point.
2 Staggered cut resulting in single-stranded / sticky ends which anneals to any complementary end from other DNA molecules cut up by the same enzyme.

(d) The plasmids are mixed with bacteria. Some bacteria took up the plasmids. Explain how it is possible to distinguish between bacteria which have taken up a plasmid with the insulin gene and those which have taken up a plasmid without any insulin gene. [4]

1 Ref to using antibiotic resistance selection as there are more than 1 antibiotic resistance gene in the plasmid
2 Ref to using nutrient agar plates with ampicillin and tetracycline
3 Ref to using bacteria that are transformed can grow on ampicillin medium as the plasmid confers the ability to breakdown ampicillin / resistance to ampicillin
4 Ref to bacteria with the recombinant plasmid / has the gene insert cannot survive on tetracycline media as the tetracycline resistant gene is disrupted

[Q3 Total: 10]

QUESTION 4
The Figure 4.1 below shows the flow of electrons in the non-cyclic and cyclic photophosphorylation.

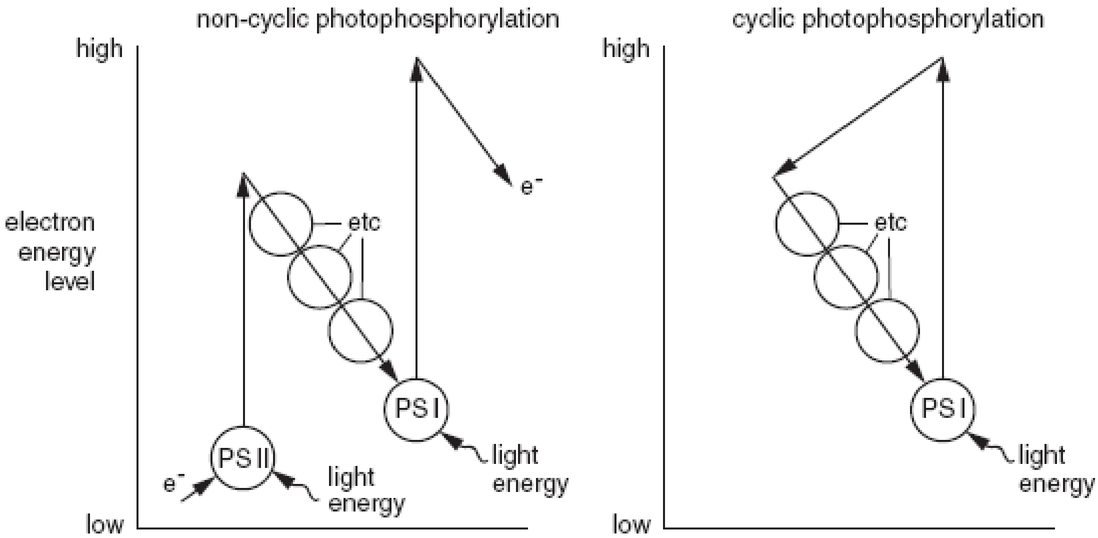


Figure 4.1

(a) State the precise location of:

(i) **Photophosphorylation** – thylakoid membrane [1/2]
(ii) **oxidation of NADPH** – stroma [1/2]

(b) Describe how non-cyclic photophosphorylation differs from cyclic photophosphorylation. [3]

Non-cyclic photophosphorylation	Cyclic photophosphorylation
Occurs when plants require ATP and NADPH.	Occurs when plants require only ATP.
Non-cyclic pathway of electrons	Cyclic pathway of electrons
Involves PS I and PS II.	Involves only PS I.
Water is the first electron donor.	PS I is the first electron donor.
$NADP^+$ is the final electron acceptor.	PS I is the final electron acceptor.
High H^+ ion concentration in thylakoid space is due to photolysis of water and active transport of H^+ ions using energy generated from the ETC.	High H^+ ion concentration in thylakoid space is due only to active transport of H^+ ions using energy generated from the ETC.
Products = ATP, NADPH and oxygen.	Product = only ATP.

(c) Briefly describe the main stages of the dark reaction. [4]

1. CO_2 accepted by RuBP (catalyzed by **RUBISCO** / RuBP carboxylase);
2. 6C intermediate breaks down /splits immediately into 2 X GP (3C compounds)
3. Each GP is phosphorylated using ATP and reduced by NADPH
4. To form 3C TP/ 3C PGAL;
5. TP is used to regenerate RuBP;
6. TP also enters different pathways to form either carbohydrates / proteins / lipids depending on plant's needs;

(d) Explain briefly what initially happens to the concentration of RuBP and GP if the supply of carbon dioxide is reduced. [2]

1. <u>**RuBP**</u>, accumulates / goes up due to reduced combination / carbon fixation with CO_2;
2. <u>**GP**</u>, goes down / not as much being formed and also due to conversion to TP;

[Q4 Total: 10]

Section B

5 (a) Describe the unique features of zygotic stem cells, embryonic stem cells and blood stem cells and explain their normal functions in a living organism. **[12]**

<u>General properties</u>

1 Self-renewal: Capable of dividing / proliferating and renewing themselves for long periods
2 Unspecialized: Do not have any tissue-specific structure that allow it to perform specialized functions
3 Able to differentiate into any mature cell type depending on the level of potency of the stem cell

<u>Zygotic stem cells</u>

4 Totipotent
5 Ref to cells in the morula (first few divisions after fertilization / fusion of an egg and sperm)
6 Able to differentiate into any cell type <u>to form whole organisms</u> / all the tissues of the body
7 Their daughter cells are pluripotent and multipotent
8 E.g. embryonic and extra-embryonic tissues

<u>Embryonic stem cells</u>

9 Pluripotent
10 Ref to e.g. inner cell mass of the blastocyst
11 They have the ability to differentiate into almost any cell type <u>to form any organ or type of cell</u>
12 Their daughter are not totipotent but are multipotent
13 E.g. any tissue of the body except the placenta
14 Important as they give rise to multiple specialized cell types that make up the heart, lung, skin and other tissues in the developing foetus

<u>Blood stem cells</u>

15 Multipotent
16 They are able to differentiate into a <u>limited range of cell type</u> / produce only cells of a closely related family of cells
17 Their daughter cells are not pluripotent or totipotent
18 E.g. Haematopoietic stem cells which can give rise to red blood cells, white blood cells and platelets
19 Important for daily blood turnover and fighting infections

(b) Describe how mitosis maintains genetic stability and its importance in growth, repair and asexual reproduction. **[8]**

1 Mitosis forms 2 daughter nuclei
2 Which are genetically identical to the parent cell
3 With the same number of chromosomes as the parent cell

4 Synthesis (S) phase of interphase, amount of DNA is doubled
5 Semi-conservative replication ensures that all genetic information is retained
6 Crossing over or pairing up of homologous chromosomes do not occur
7 Ensures that there is no genetic variation / maintains genetic stability

Growth
8 Mitosis plays a role in the growth in tissues as increase in cell numbers results in increase in size of organisms (growth) in multicellular organisms
9 New cells are identical with existing cells so that they carry out the same function

Repair
10 Mitosis is important for the repair / replacement of worn-out parts of the body
11 Important for tissues that replaced damaged cells with exact copies of the original cells in order for the tissue to function properly

Asexual reproduction
12 Mitosis forms the basis of asexual reproduction as the separated cells / plant parts become new offspring
13 Forms clones that are genetically identical with parents and thus facilitates successful colonization of habitats by species

6 (a) Describe the structure and roles of ribosome and transfer RNA in the synthesis of proteins. **[10]**

<u>Ribosome</u>

1 40S / Small subunit holds mRNA at 5' ends OR reads codons on mRNA
2 60S / Large subunit has 2 sites for translation of codons into amino acid sequenes
3 'P' Site holds first amino acid-tRNA complex which recognizes the start codon
4 Initiate polypeptide chain formation
5 Catalytic site catalyses formation of peptide bond between amino acids.
6 'A' Site holds in place the amino acid-tRNA complex next in sequence
7 Site of elongation of polypeptide chain
8 Polyribosomes / many ribosomes may move along the mRNA at the same time, forming many identical polypeptide → increase the rate of translation.

<u>tRNA</u>

9 Carries amino acid to the ribosome during protein synthesis
10 CCA site at the 3' end allows for the attachment of a specific amino acid
11 Has a specific anti-codon that is complementary with codon on mRNA
12 This allows sequence of codons on mRNA to be translated into the correct sequence of amino acids in the polypeptide.

(b) Using a named example, explain how natural selection may bring about evolution. **[10]**

1 Ref to a named example: Galapagos finches / *Biston besularia* Moth
2 Populations have great reproductive potential
3 But numbers remain approximately constant

4 As many fail to survive because of factors in the environment being selective agent or pressure
5 Ref. to named example of selection pressure: food / shelter / breeding sites / predation / diseases
6 Selection process include directional selection, stabilizing selection and disruptive selection

7 Individuals with genetic variations best adapted to the environment more likely to survive
8 Reproductive age and pass on their beneficial genes to offspring

9 Selection of advantageous phenotypes lead to <u>change in frequency</u> of particular alleles and genotypes
10 Proportion of individuals possessing the advantageous characteristic increases whereas that lacking the characteristic decreases

11 Lead to evolutionary change over a <u>long</u> period of time
12 New species arising from <u>existing</u> species / ref. speciation theory